

UMC Data Hackathon

Mentors' profiles



Viviana Umpierrez is a statistician at the International Telecommunication Union (ITU), where she specializes in ICT indicators derived from a wide range of data sources, including administrative records and household surveys. Her work contributes to advancing the global measurement of digital development and strengthening data-driven approaches in the telecommunications sector.

She brings over two decades of experience in quantitative analysis, combining statistical modelling, predictive analytics, and data visualization to support strategic decision-making. Before joining the ITU, she led the Economic Analysis team at ANTEL, Uruguay's national telecommunications provider, for nearly 14 years. There, she leveraged microdata to uncover behavioral patterns and model user dynamics, applying statistical methods to analyze consumer behaviour, market trends, and service performance across diverse contexts. Her career also includes roles at Coca-Cola, Tata Consultancy Services, and academic research in macroeconomics at Universidad de la República.

Viviana holds a Bachelor's degree in Economics from Universidad de la República and a Master's degree in Economics from Universidad de Montevideo, both in Uruguay. Her technical expertise spans statistical programming, data integration, and the application of analytical frameworks to complex datasets in both public and private sector environments.



José L. Cervera-Ferri, Senior Project Manager at the ICT Data and Analytics Division of the International Telecommunication Union (ITU) has more than 30 years of experience in governmental and private institutions including all the different phases of statistical production, dissemination and planning.

José got a MSc in Mathematics (Statistics and Operations Research) from the University of Valencia, Spain. Senior Statistician of the Spanish administration since 1995, he served as member of the executive board of the Spanish Statistical and Operations Research Society (SEIO) and serves currently as Associated Editor of the Statistical Journal of the International Association for Official Statistics.

His professional experience includes statistical capacity building activities in developing and transition countries. In relation to ICT statistics, he was responsible for the design of the first household and business surveys on the use of ICT in the National Statistical Institute of Spain (INE), and is the main author of the UNCTAD *Manual for the Production of statistics on the Information Economy* and the ITU *Manual on measuring ICT access and use by households and individuals*. He has taught graduate courses on statistics for Computer Science and Civil Engineering (Universidad Politécnica de Valencia, Spain), Operations Research and Market Research Techniques (Universidad Pontificia de Comillas, Madrid, Spain); courses for official statisticians and data users, as well as on Machine Learning within the European Statistical Training Programme, addressed to statisticians of the National Statistical Institutes of the EU and Eurostat.



Esperanza Magpantay is the Senior Statistician at the ICT Data and Analytics Division of the International Telecommunication Union (ITU), where she has worked since 2000. She leads the global collection, harmonization, and dissemination of ICT statistics and spearheaded the creation of the ITU Expert Groups on ICT Indicators, which have become key global mechanisms for developing methodologies and standards in ICT measurement.

She represents ITU in major international forums, including the UN Statistical Commission, the UN World Data Forum, and the Partnership on Measuring ICT for Development.

Esperanza also chairs the *UN Committee of Experts on Big Data and Data Science for Official Statistics' Task Team on Mobile Phone Data*, a global collaboration of over 50 experts advancing the use of mobile data for official statistics. She co-leads the ITU-World Bank project Partnership to Put Mobile Phone Data to Work for Policy and Statistics, which supports countries in using mobile phone data to inform public policy and its use for official statistics.

Her expertise includes estimation and forecasting, advanced analytics using machine learning and data science, and proficiency in data storytelling, visualization, and dissemination. With over three decades of experience in international statistics, Esperanza has been a leading advocate for improving information society statistics, supporting countries in producing high-quality official data, and fostering collaboration and coordination among national and international stakeholders to strengthen evidence-based policymaking. She has led the development and production of international methodological guides, delivered capacity-building workshops and organized global events, and contributed to shaping global frameworks and digital indicators that strengthen the production and use of official data worldwide.

Prior to joining ITU, she worked as a Statistical Officer at the International Labour Office (ILO) for seven years. She holds a *Global Master in Business Analytics and Big*

Data from IE University in Madrid, Spain, and a *Bachelor of Science in Statistics* from the University of the Philippines. Passionate about innovation and collaboration, Esperanza continues to champion the use of technology and data to bridge the digital divide and support evidence-based policymaking.



Pranita Upadhyaya is a results-oriented technology expert with over 28 years of experience across ICT, data science, and digital transformation. Her expertise spans IT management, software engineering, telehealth, and the development of AI-powered disease prediction models, driving innovation in healthcare, education, and governance.

Currently serving as Head, ITU Area Office and Innovation Centre for South Asia at the International Telecommunication Union (ITU), she drives regional ICT initiatives aligned with global priorities and the Sustainable Development Goals (SDGs). She builds strategic partnerships with governments, industry, and regional bodies across South Asia to promote digital inclusion, resource mobilization, and scalable innovation.

Her data science acumen is evident in her pioneering work in predictive analytics and AI integration within healthcare systems. Prior to joining ITU, as a Chief Technical Officer at the B.P. Eye Foundation, she led the deployment of advanced hospital management systems, developed a data-driven business forecast report leveraging machine learning and statistical modelling to predict financial and operational outcomes, strengthened cybersecurity frameworks, and introduced telemedicine and AI-powered diagnostic services that transformed healthcare accessibility in underserved communities.

In academia, Pranita brings over 28 years of combined full-time and part-time experience. She served as Vice Principal and Professor at Kathford International College of Engineering and Management, where she led curriculum innovation, enhanced digital learning engagement, and modernized IT infrastructure.

Her career spans UN, government, NGOs, academia, and the private sector, with a focus on leveraging emerging technologies for social impact. With a PhD in Computer Engineering and publications in cybersecurity, e-government, and AI, she is recognized for her contributions to digital transformation and data science.



Fredrik Eriksson is a Senior Data Scientist of the ICT Data and Analytics division of the International Telecommunication Union (ITU) with 20 years of experience in quantitative analysis from international organisations and across the private sector. Fredrik has previously been responsible for ITU's data collection on ICT household statistics, and re-joined ITU in 2021 to focus on ITU's work on big data and geospatial analysis after spending a few years building a Data Science team in the private sector.

Prior to first joining ITU, he worked in the finance sector and spent eight years at the OECD analysing international finance flows for economic development. Fredrik holds a Master's degree in Statistics from Uppsala University in Sweden.



Akanksha Sharma is a Programme Officer at the International Telecommunication Union's Area Office for South Asia & Innovation Centre in New Delhi, serving as the regional focal point for digital innovation in the Asia Pacific. She leads ITU's programmes and projects on emerging technologies, particularly artificial intelligence, and oversees regional initiatives on e-waste management for the region.

Previously, Akanksha worked as a Senior Policy Officer at the Netherlands Innovation Network in Bangalore, where she facilitated innovation and technology cooperation between India and the Netherlands across sectors such as high-tech systems (including space), life sciences & healthcare, and key enabling technologies like AI, IoT, and big data. She worked closely with government, industry, and academia in both countries to foster ecosystem connections, drive co-creation, and develop innovative, sustainable solutions for shared societal challenges. With over 15 years of experience in building public-private partnerships and leveraging technology and innovation to address societal needs, Akanksha brings deep expertise in multi-stakeholder engagement, programme management, and the application of technology to advance the Sustainable Development Goals.

She holds a bachelor's degree in computer science engineering, an MBA in Marketing & Finance, and a certification in *Innovation & Change Leadership* from

the Indian School of Business, Hyderabad.



Nathan Menton is a statistician in the ITU's ICT Data and Analytics Division. His work focuses on the measurement of ICT indicators through household surveys, including the production and estimation of key indicators such as individual Internet use and mobile phone ownership. He manages ICT data collection through country questionnaires and contributes to the statistical methods underpinning ITU's Facts & Figures publication.

Nathan also works on the development and revision of international statistical recommendations on topics such as digital skills and AI use, primarily through the ITU's Expert Group on ICT Household Indicators (EGH). His role includes supporting methodological updates, integrating new indicators into ITU data systems, and collaborating with national statistical offices to improve data quality and comparability. Before joining ITU, he worked as a statistician at the United Nations Economic Commission for Europe, the United Nations Statistics Division, and the United Nations Office at Geneva. Earlier in his career, he worked in statistical and analytical roles in the United States, including survey research and cost and risk analysis. He holds a master's degree in statistical science from George Mason University.



Nehal Desai is a Statistical assistant at the ICT Data and Analytics Division of International Telecommunication Union (ITU), working on the data collection, harmonization and dissemination of ICT statistics related to telecommunications infrastructure, household ICT access and use, and ICT price indicators. She works closely with regulators and ministries, national statistical offices to strengthen the robustness, consistency, and international comparability of ICT data, supporting evidence-based digital development policies.

She contributes to the enhancement of international statistical frameworks and to the integration of new indicators and methodologies through her involvement in the ITU's Expert Groups on ICT indicators.

Prior to joining ITU, she worked as a Database Administrator in a Swiss bank in the private sector. Nehal holds a master's degree in mathematics from Gujarat University in India.



Amarnath Ojha is presently serving as Director at the National Communications Academy-Finance (NCA-F), the apex training institution of the Government of India for finance, policy, and capacity building in the telecommunications sector. He is a serving officer of the Indian Posts & Telecommunications Accounts and Finance Service (IPTAFS), 2018 batch, and brings over eleven years of professional experience spanning both the private and public sectors.

He holds a Bachelor's degree in Mechanical Engineering and a Master's degree in Public Administration, combining technical grounding with an understanding of public governance and institutional systems.

Within the Government of India, he has held diverse assignments in public finance, administration, training, and project management. He served at the Principal Controller of Communications Accounts Office, Hyderabad, where he worked on revenue assessment, USOF, and pension disbursement, contributing to financial oversight and service delivery mechanisms. At NCA-F, he has played an active role in academic planning and institutional capacity building, including serving as the Focal Point Officer for the implementation of competency-based capacity building initiatives under Mission Karmayogi from 2022 to 2024.

He has served as Course Director for Trainer Development Programs and ITEC international training programmes, and as Course Coordinator for Officer Trainees' induction training, MCT training, and several national and international capacity-

building initiatives. He is a Recognised Lead Trainer and Master Trainer for the Rashtriya Karmayogi Large Scale Jan Seva Program.

Currently, he is serving as Director (Project) for the development of the new NCA-F campus, coordinating with multiple stakeholders. His work focuses on institutional development, inter-agency coordination, and strengthening training infrastructure to support future-ready capacity building in the communications sector.



Sh. Shashank Shekhar Agarwal is presently serving as Deputy Director in the Department of Telecommunications (DoT), Government of India. A technical and policy professional, he has been associated with the telecom sector since August 2018 and has been actively contributing to policy formulation, regulatory implementation, and technology-driven governance in India's rapidly evolving communications landscape.

An alumnus of the Indian Institute of Technology (IIT) Roorkee, he holds a Bachelor of Technology (B.Tech.) degree, which has provided him with a strong foundation in engineering systems, analytical thinking, and problem-solving. His academic background enables him to effectively bridge technical considerations with public policy and regulatory frameworks.

In his professional role at the Department of Telecommunications, Sh. Agarwal is involved in matters relating to telecom policy, regulatory compliance, and the strengthening of digital and telecommunications infrastructure. His work supports initiatives aimed at enhancing service delivery, improving regulatory efficiency, and enabling digital transformation in the communications sector.

He possesses hands-on technical proficiency in tools and programming languages such as SolidWorks, C++, and HTML, which complement his responsibilities in technology management and policy analysis. His areas of expertise include telecommunications policy and regulation, digital infrastructure, engineering systems analysis, and software-oriented problem solving. He has a working

knowledge of data analytics and its application in policy evaluation, regulatory monitoring, and informed decision-making, particularly in the context of telecommunications and digital governance. With a blend of technical knowledge and policy experience, he continues to contribute meaningfully to India's telecom governance ecosystem.